A FLAT FINITE TYPE MORPHISM WITH GEOMETRICALLY INTEGRAL FIBRES IS UNIVERSALLY PURE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Lifting of specializations along a flat morphism. Morphisms, label morphisms-lemma-generalizations-lift-flat, invoked at proof line 5137. Lets the specialization in the base be lifted to a specialization in the source ending at the generic point of the special fibre, which rules out an impurity.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Navigation pointer for \ref{morphisms-lemma-generalizations-lift-flat} used inside the counted span (line 5137); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a finite type, flat morphism of schemes with geometrically integral fibres. Then X is universally pure over S.*

Proof. Let $\xi \in X$ with $s' = f(\xi)$ and $s' \rightsquigarrow s$ a specialization of S. If ξ is an associated point of $X_{s'}$, then ξ is the unique generic point because $X_{s'}$ is an integral scheme. Let ξ_0 be the unique generic point of X_s. As $X \to S$ is flat we can lift $s' \rightsquigarrow s$ to a specialization $\xi' \rightsquigarrow \xi_0$ in X, see Morphisms, Lemma 03HV. The $\xi \rightsquigarrow \xi'$ because ξ is the generic point of $X_{s'}$ hence $\xi \rightsquigarrow \xi_0$. This means that $(\mathrm{id}_S, s' \to s, \xi)$ is not an impurity of $\mathcal{O}_X$ above s. Since the assumption that f is finite type, flat with geometrically integral fibres is preserved under base change, we see that there doesn't exist an impurity after any base change. In this way we see that X is universally S-pure. □

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.